

Subject Description Form

Subject Code	EE512
Subject Title	Electric Vehicles
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To acquire a broad knowledge on modern electric vehicles (EVs). 2. To understand the development of EVs from technological, environmental, and societal perspectives.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the importance of EVs for environment, energy sustainability and climate change. b. Understand various underpinning technologies for modern EVs, including electric motor drives, energy storage, batteries, charging methods, infrastructure and auxiliary systems. c. Explain the emerging technologies such as hybrid electric vehicles (HEVs), fuel cell electric vehicles (FEV) and energy storage methods.
Subject Synopsis/ Indicative Syllabus	1. Introduction to electric vehicles (EVs): Historical perspective. EV advantages and impacts. EV market and promotion: infrastructure needs, legislation and regulation, standardization. 2. Electric vehicle (EV) design options: EV configurations: fixed vs. variable gearing, single- vs. multiple-motor drive, in-wheel drives. EV parameters, driving cycles and performance specifications. Choice of system voltage levels: electrical safety and protection. 3. Vehicle dynamics and motor drives: Road load: Vehicle kinetics; Effect of velocity, Acceleration and grade. EV drivetrain and components. EV motor drive systems: DC drives, Induction motor drives, Permanent-magnet synchronous motor drives, Switched reluctance motor drives. Control strategies. 4. Batteries: Battery parameters. Types and characteristics of EV batteries. Battery testing and maintenance; Charging schemes. Battery Management System. Open-circuit voltage and ampere-hour estimation. Battery load levelling Energy Storage. 5. Auxiliaries: On-board and off-board battery chargers. Energy management units. Battery state-of-charge indicators. Temperature control units. Power steering. 6. Emerging EV technologies: Hybrid electric vehicles (HEVs): types, operating modes, torque coordination and control, generator/motor requirements. Fuel cell electric vehicles (FEVs): fuel cell characteristics, hydrogen storage systems, reformers. Alternative sources of power: super- and ultra-capacitors, flywheels.

Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures, complemented by tutorials and worked examples. Self-learning on the part of students is strongly encouraged and extensive use of web resources will be made. A term paper and a related presentation enable students to develop skills in literature survey and writing. Oral presentation sessions develop students’ skills in spoken communication and peer evaluation.				
	Teaching/Learning Methodology	Outcomes			
		a	b	c	
	Lectures	√	√	√	
	Tutorials	√	√	√	
	Assignment and oral presentation	√	√	√	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Examination	50%	√	√	√
	2. Test	30%	√	√	√
	3. Assignment (Term Paper/mini project/ Homework)	20%	√	√	√
	Total	100%			
	It is an advanced elective on electric vehicles. The outcomes on electric vehicle technology and its impacts are assessed by the usual means of test and examination, and partly by the term paper. The outcomes on technical communication and presentation skills are evaluated by the term paper and a related oral presentation.				
Student Study Effort Expected	Class contact:				
	▪ Lecture/Tutorial			30 Hrs.	
	▪ Presentation/Tests			9 Hrs.	
	Other student study effort:				
	▪ Self-study and revision			48 Hrs.	
	▪ Report – Case Study			18 Hrs.	
	Total student study effort			105 Hrs.	
Reading List and References	Reference books: 1. David Bricknell, Electric Vehicle Technologies, 2020. 2. K.T.Chau, Energy Systems for Electric and Hybrid Vehicle, IET, Aug 2016 3. Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals, New York: CRC Press, 2nd edition, 2010. 4. Per Enge, Nick Enge, Stephen Zoepf, Electric Vehicle Engineering, McGraw Hill, 1st Edition, 2020. 5. Dharavath Kishan, Ramani Kannan, B Dastagiri Reddy, Prajof Prabhakaran, Power Electronics for Electric Vehicles and Energy Storage: Emerging Technologies and Developments, CRC Press, 2023				